Daily Geometry Drill #4

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Exterior angles sum to 360° *· Isosceles chains · Regular polygons as equal triangles · Similar triangles · Area ratios.*

Section A Rapid Recognition

1. Write down the size of each interior angle of a regular 12-sided polygon.

Ans: 150°

2. An isosceles triangle has apex angle 40°. Write down each base angle.

Ans: 70°

3. The incircle of the right triangle with legs 3 and 4 touches all three sides. What is the inradius?

Ans: 1

4. Each interior angle of a regular polygon is 162°. Write down the number of sides.

Ans: 20

5. Write down the number of diagonals of a decagon.

Ans: 35

6. The angle bisector from A of triangle ABC meets BC at D, with $AB : AC = 2 : 3$ and $BC = 10$. Write down DC.

Ans: 6

7. The isosceles triangle with sides 5, 5, 6 has base 6. Write down the median from the apex to the base.

Ans: 4

8. A median of a triangle has length 12. How far is the centroid from the vertex at the end of that median?

Ans: 8

9. Two similar triangles have corresponding sides in the ratio $2:3$. The smaller has area 8. Write down the area of the larger.

Ans: 18

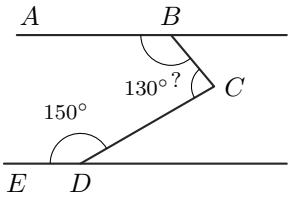
10. Write down the area of a regular hexagon with side 2.

Ans: $6\sqrt{3}$

Section B Manipulation Drills

1. AB is parallel to ED. Given $\angle ABC = 130°$ and $\angle CDE = 150°$, find $\angle BCD$.

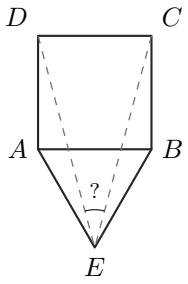
A) 60°
B) 70°
C) 80°
D) 100°
E) 110°



Ans: C) 80°

2. $ABCD$ is a square and ABE is an equilateral triangle drawn outside the square on side AB. What is $\angle DEC$? *(after Primer 1.4 P11)*

A) 15°
B) 20°
C) 45°
D) 60°
E) 30°



Ans: E) 30°

3. The triangle with sides 13, 14, 15 has area 84. Its inscribed circle has what radius?

Ans: 4

4. The five diagonals of a regular pentagon form a five-pointed star whose points are the pentagon's vertices. What is the angle at each point of the star?

A) $36°$

B) $30°$

C) $45°$

D) $54°$

E) $72°$

Ans: A) $36°$

5. $ABCDEF$ is a regular hexagon. What fraction of its area is triangle ACE?

A) $\frac{1}{3}$

B) $\frac{1}{2}$

C) $\frac{2}{3}$

D) $\frac{3}{4}$

E) $\frac{1}{4}$

Ans: B) $\frac{1}{2}$

6. A square, a regular hexagon and one other regular polygon fit together around a point with no gaps or overlaps. How many sides does the other polygon have?

A) 8

B) 10

C) 15

D) 12

E) 18

Ans: D) 12

7. Each interior angle of a regular polygon is four times each exterior angle. How many sides does it have?

A) 10

B) 8

C) 9

D) 12

E) 16

Ans: A) 10

8. $ABCDEFGH$ is a regular octagon. Find $\angle ACF$.

Ans: 67.5°

9. A regular pentagon and a regular hexagon share a side and lie on opposite sides of it. At each end of the shared side there is a gap between the two shapes. What is the angle of this gap?

A) 120°

B) 126°

C) 128°

D) 132°

E) 144°

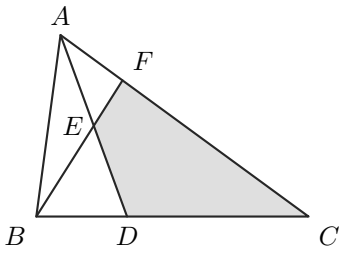
Ans: D) 132°

10. A five-pointed star is drawn with five straight strokes, not necessarily regular. Write down the sum of the angles at its five points.

Ans: 180°

Section C Substitution & Structure

1. In triangle ABC, the point D lies on BC with $BD : DC = 1 : 2$, and E is the midpoint of AD. The line BE meets AC at F. What fraction of triangle ABC is the shaded quadrilateral $EDCF$?



A) $\frac{1}{2}$

B) $\frac{5}{8}$

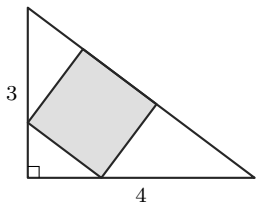
C) $\frac{7}{12}$

D) $\frac{2}{3}$

E) $\frac{3}{4}$

Ans: C) $\frac{7}{12}$

2. A right-angled triangle has legs 3 and 4. A square sits inside it with one side along the hypotenuse and its other two vertices on the legs. What is the side length of the square?



A) $\frac{12}{7}$

B) $\frac{5}{3}$

C) $\frac{3}{2}$

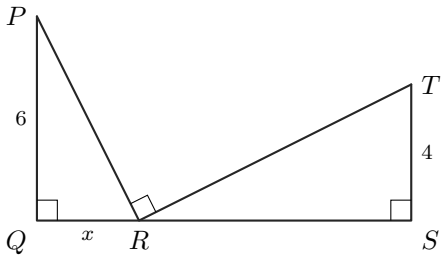
D) $\frac{60}{37}$

E) $\frac{60}{49}$

Ans: D) $\frac{60}{37}$

3. PQ and TS are perpendicular to QS, with $PQ = 6$, $TS = 4$ and $QS = 11$. The point R lies on QS with $QR = x$, and $\angle PRT = 90°$. Then *(after Primer 1.4 P48)*

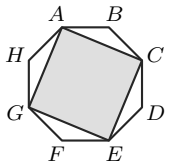
A) x has two possible values whose sum is 12
B) x has two possible values whose difference is 5
C) x has only one value, and $x \geq 5.5$
D) x has only one value, and $x < 5.5$
E) x cannot be determined from the given information



Ans: B) x has two possible values whose difference is 5

4. $ABCDEFGH$ is a regular octagon with sides of length 1. What is the area of the shaded square $ACEG$?

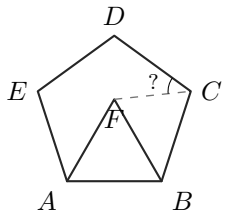
A) $2 + \sqrt{2}$
B) $1 + \sqrt{2}$
C) $2 + 2\sqrt{2}$
D) 4
E) $3 + 2\sqrt{2}$



Ans: A) $2 + \sqrt{2}$

5. $ABCDE$ is a regular pentagon, and ABF is an equilateral triangle with F inside the pentagon. What is $\angle FCD$?

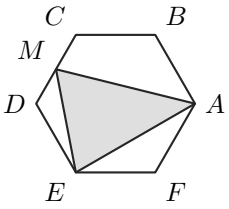
A) $36°$
B) $48°$
C) $42°$
D) $54°$
E) $66°$



Ans: C) 42°

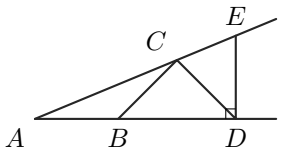
6. $ABCDEF$ is a regular hexagon and M is the midpoint of CD. What fraction of the hexagon is the shaded triangle AME?

A) $\frac{1}{3}$
B) $\frac{1}{2}$
C) $\frac{1}{4}$
D) $\frac{3}{8}$
E) $\frac{5}{12}$



Ans: E) $\frac{5}{12}$

7. The points B and D lie on one arm of an angle at A, and C and E lie on the other arm, with $AB = BC = CD = DE$. The line DE is perpendicular to AD. What is $\angle BAC$?



A) $15°$
B) $18°$
C) $20°$
D) $22.5°$
E) $30°$

Ans: D) $22.5°$

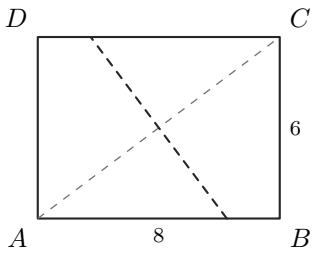
8. A regular 12-sided polygon is inscribed in a circle of radius 1. What is the total area inside the circle but outside the polygon?

A) $\pi - 2$
B) $\pi - 3$
C) $\pi - \dfrac{3\sqrt{3}}{2}$
D) $\pi - 2\sqrt{2}$
E) $\dfrac{\pi}{12}$

Ans: B) $\pi - 3$

Section D Challenge

1. A rectangular sheet $ABCD$ has $AB = 8$ and $BC = 6$. It is folded along a straight crease so that C lands exactly on A. How long is the crease?



A) 7

B) $5\sqrt{2}$

C) $\dfrac{25}{4}$

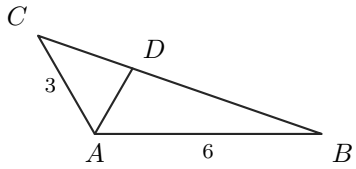
D) 8

E) $\dfrac{15}{2}$

Ans: E) $\frac{15}{2}$

2. In triangle ABC, $AB = 6$, $AC = 3$ and $\angle BAC = 120°$. The bisector of angle A meets BC at D. What is the length of AD?

A) $2\sqrt{3}$
B) 2
C) 4
D) $\sqrt{7}$
E) 3



Ans: B) 2

3. What is the side length of the largest square that fits inside a regular hexagon with sides of length 1?

A) 1

B) $\dfrac{2\sqrt{3}}{3}$

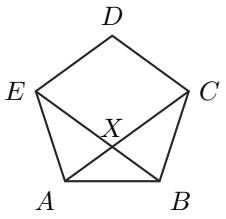
C) $3 - \sqrt{3}$

D) $\dfrac{\sqrt{6}}{2}$

E) $\sqrt{2}$

Ans: C) $3 - \sqrt{3}$

4. $ABCDE$ is a regular pentagon with sides of length 1. The diagonals AC and BE meet at X. What is the length AX?



A) $\dfrac{\sqrt{5}-1}{2}$

B) $\dfrac{\sqrt{5}+1}{2}$

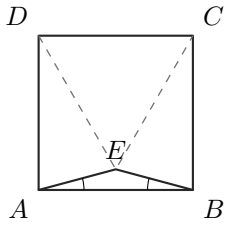
C) $\dfrac{1}{2}$

D) $\dfrac{\sqrt{5}}{2}$

E) $\dfrac{3-\sqrt{5}}{2}$

Ans: A) $\frac{\sqrt{5}-1}{2}$

5. $ABCD$ is a square. The point E inside the square satisfies $\angle EAB = \angle EBA = 15°$. Prove that triangle CDE is equilateral.



Ans: Proof: see method